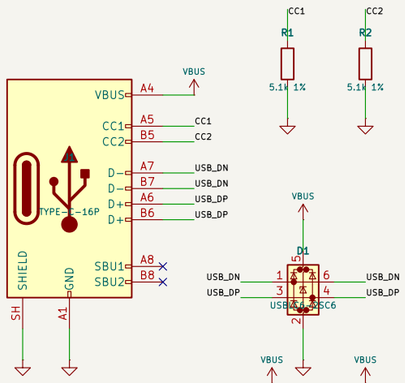
1. USB-C 5V INPUT & ESD PROTECTION

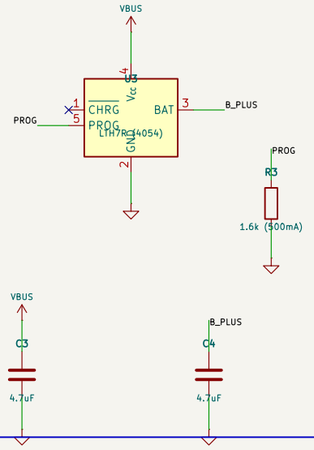
DOST EPDC: 1.0mm round PTH shield holes



2. LI-ION CHARGER (LTH7R / LTC4054)

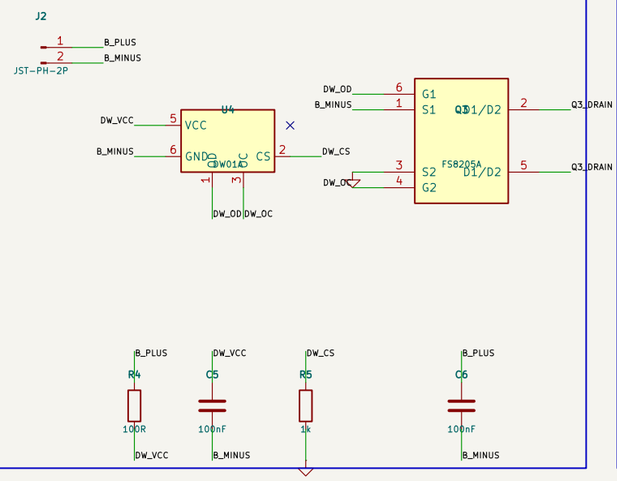
CC/CV 4.2V: I_prog = 1000V / R_prog

Pin 1: CHRG status LED (DNP)
Pin 5: R_prog=1.6k sets 500mA fast charge



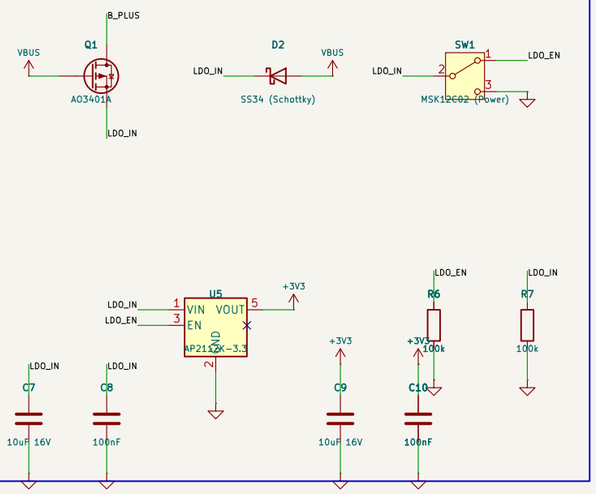
3. BATTERY PROTECTION (DW01A + FS8205A)

Fixed Rev A TSSOP defect: True SOT-23-6 dual NMOS



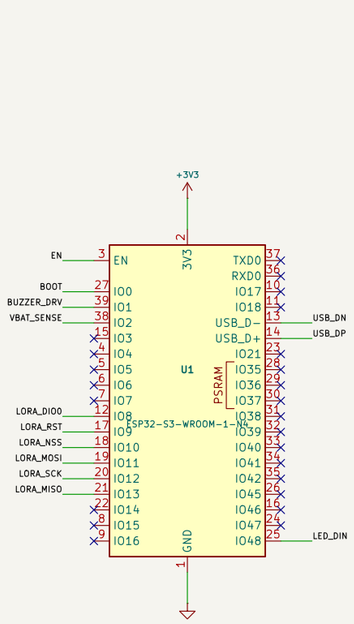
4. POWER PATH MANAGEMENT & 3.3V LDO

Seamless USB/Battery handoff + ultra-low-noise 600mA LDO



5. ESP32-S3-WROOM-1 MCU SUBSYSTEM

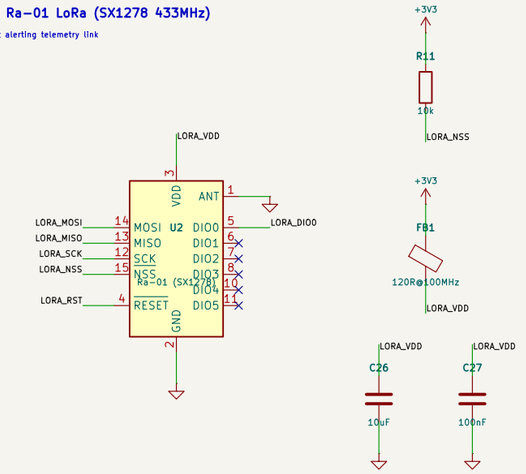
DOST EPDC compliant: 0.40mm drill thermal vias, native USB, LoRa SPI bus



GPIO19 = USB D-, GPIO20 = USB D+ (Native USB OTG)
GPIO48 = NeoPixel RGB LED Data (Rev A fix)
GPIO35–37 reserved for on-module Octal PSRAM
On-module 40MHz crystal & antenna included

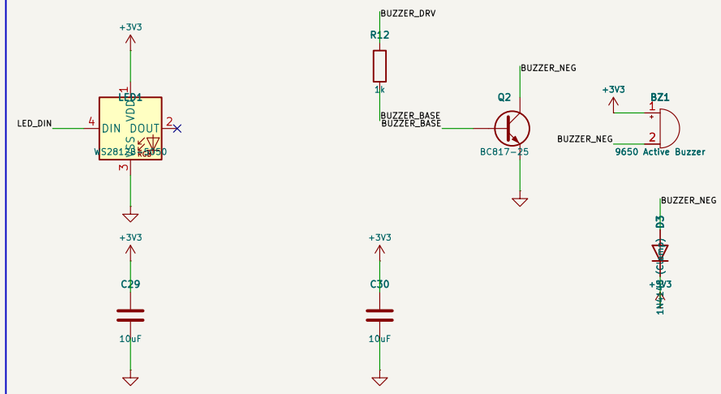
6. AI-Thinker Ra-01 LoRa (SX1278 433MHz)

Long-range accident alerting telemetry link



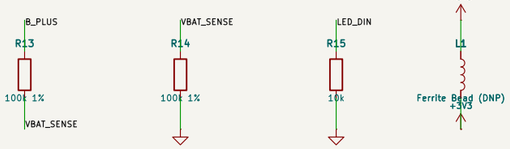
7. PERIPHERALS: WS2812B RGB LED & BUZZER

Status alerts + audible danger alarm



8. BATTERY TELEMETRY & MISC DNP

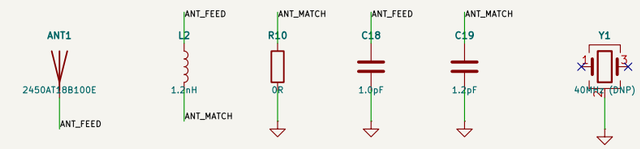
ADC divider: V_sense = V_bat * R14 / (R13 + R14)



Divider ratio: 1/2. Max 4.2V -> 2.1V ADC input.
L1: Optional high-frequency filter (DNP).

9. BLE CHIP ANTENNA & 40MHz XTAL

Preserved for Rev A schematic parity; on-module in WROOM-1



Replaces defective Rev A; integrates ESP32-S3-WROOM-1, Ra-01 LoRa, and DW01A+FS8205A
Target: DOST-ASTI EPDC 2-Layer FR4 (0.30mm trace, 0.254mm clearance, 0.40mm drill)
68 components, 38 nets, 9 functional blocks -- strictly source-traceable

Antigravity / Jan Clyde T. Talosig

Sheet: /
File: RAMS_RevC.kicad_sch

Title: RAMS Wearable Node -- Master Schematic

Size: A3	Date: 2026-10-30	**Rev: Rev C (DOST-ASTI EPDC**
KiCad E.D.A. 10.0.6		Id: 1/1